

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

Data Science Lab (DSL) — Course Project Report

Credit Risk Modelling:

An Explainable Credit Scoring System: Integrating Default Probability Estimation with CIBIL Score Calculation and Borrower Guidance

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This report presents a comprehensive machine-learning-driven credit risk modelling framework designed for deployment as an internal credit decisioning tool within Indian banks and Non-Banking Financial Companies (NBFCs). The system addresses critical gaps in traditional credit assessment, including shallow explainability, rigid decision thresholds, and the absence of calibrated default probabilities.

Using the "Give Me Some Credit" dataset (150,000 borrower records with 11 financial attributes), we train and evaluate four model architectures: Logistic Regression (baseline), XGBoost, CatBoost (primary production model), and a PyTorch Mixture-of-Experts (MoE) neural network. CatBoost, optimised via Optuna Bayesian hyperparameter search (50 TPE trials across 7 parameters), achieves a ROC-AUC of approximately 0.86 on the held-out test set. A dual-threshold decision framework (Max-F1 and Precision-targeted) enables banks to calibrate precision-recall trade-offs to their specific risk appetite.

Each prediction is accompanied by a SHAP-based explainability layer providing per-feature impact quantification and actionable improvement recommendations. The model output is further transformed into a CIBIL-aligned credit score (300–900) using a composite formula weighting default probability at 50%, payment history at 25%, credit utilisation at 15%, and credit mix at 10%. The system is deployed via a Streamlit dashboard for real-time scoring and a Google Sheets API integration for batch processing.

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1. Introduction

Credit risk assessment remains one of the most consequential decision-making processes in the financial services industry. In India, the banking sector disbursed over INR 150 lakh crore in credit by FY 2024–25, yet Non-Performing Assets (NPAs) continue to pose systemic threats. The Reserve Bank of India reported gross NPAs of approximately 2.8% across scheduled commercial banks as of March 2025, translating to billions of rupees in potential losses.

1.1 The Present Gap

The current credit risk landscape in India is characterised by several structural deficiencies:

- **Over-reliance on static scorecards:** Most Indian banks and NBFCs employ rule-based or logistic-regression-driven scorecards that assign fixed weights to a limited set of financial variables. These models struggle to capture non-linear interactions between features such as revolving credit utilisation, delinquency history, and income volatility.
- **Opacity in credit decisions:** Borrowers denied credit rarely receive clear, actionable explanations of why they were rejected or what steps could improve their eligibility.
- **Reactive rather than predictive posture:** Traditional approaches assess creditworthiness at a point in time, rather than modelling the probability of future default. A borrower who is currently solvent may exhibit behavioural patterns (e.g., increasing delinquency frequency) that strongly predict future distress.
- **Threshold rigidity:** Conventional models apply a single decision boundary. In practice, different business contexts — secured vs. unsecured lending, retail vs. SME portfolios — demand different precision-recall trade-offs.

This project addresses these gaps by developing a machine-learning-driven credit risk modelling framework that combines predictive accuracy with explainability, offering both a probability of default (PD) and a CIBIL-aligned credit score enriched with SHAP-based rationale for every prediction.

2. Problem Definition

The core problem is to predict the probability of default: given a set of financial and demographic attributes of a borrower, calculate a CIBIL score based on a weighted average of the probability of default and other financial features, and provide actionable suggestions to increase the borrower's CIBIL score.

2.1 Formal Statement

The objective is to learn a function $f: X \rightarrow [0, 1]$ that estimates $P(Y = 1 | X)$, the probability of default, and then transform this probability into a CIBIL-aligned credit score (300–900) using a composite weighted formula. For each prediction, SHAP-based explanations identify the key risk drivers and generate targeted recommendations for the borrower to improve their score.

2.2 Extended Objectives

- **Credit Score Generation:** Transform the model output into a CIBIL-aligned credit score on the 300–900 scale using a composite weighted formula incorporating default probability (50%), payment history (25%), credit utilisation (15%), and credit mix (10%).
- **Explainability:** Provide per-feature SHAP decomposition for every prediction, identifying risk drivers and protective factors with actionable improvement recommendations.
- **Deployment-Ready Pipeline:** Deliver a production-grade system comprising a modular Python pipeline, a Streamlit dashboard for real-time scoring, and Google Sheets integration for batch predictions.

2.3 Constraints

- **Class imbalance:** ~6.7% default rate (approximately 14:1 class ratio), requiring class weighting and threshold calibration.
- **Missing data:** Monthly income missing for ~19.8% of records; dependents for ~2.6%. Handled via median imputation with missingness-as-feature indicators.
- **Sentinel values:** Delinquency fields contain codes 96 and 98 (data-entry artefacts, not actual counts), requiring special treatment.

3. Value Proposition

This project delivers a comprehensive internal credit decisioning tool for banks and NBFCs. The value proposition spans three dimensions:

3.1 Predictive Superiority

By leveraging gradient-boosted decision trees (CatBoost) with Optuna hyperparameter optimisation and a neural Mixture-of-Experts architecture, the system achieves a ROC-AUC of approximately 0.86 on the held-out test set — a meaningful improvement over the logistic regression baseline (~0.82). The MoE model captures subpopulation heterogeneity through expert specialisation, where different experts learn to handle distinct borrower segments.

3.2 Explainability and Trust

Every prediction is accompanied by a full SHAP waterfall decomposition via . Loan officers see exactly which features drive risk (e.g., high revolving utilisation, recent 90+ day delinquency) and which are protective (e.g., stable income, low debt ratio). This satisfies regulatory expectations around model governance and fair lending.

3.3 Dual Threshold Strategy

The system offers two calibrated threshold strategies rather than a single cut-off:

- Strategy A (Max-F1): Optimises the balance between precision and recall on the default class, suitable for general portfolio screening.
- Strategy B (Precision $\geq$ 50%): Maximises recall subject to the constraint that default-class precision remains at or above 50% — ideal for conservative lending environments.

4. Market Segmentation

India's total credit outstanding exceeded INR 170 lakh crore by early 2026. The Credit Information Bureau market is projected to grow at a CAGR of ~14% through 2028.

4.1 Key Segments

Segment	Characteristics	Credit Risk Challenge
Public Sector Banks	Large legacy portfolios, compliance-heavy	Manual underwriting, slow model refresh, high NPA exposure
Private Banks	Technology-forward, data-rich, growing retail books	Need for real-time scoring and personalised risk assessment
NBFCs / HFCs	Niche lending (microfinance, housing, vehicle)	Thin-file borrowers, limited bureau data, high delinquency variance
Fintech Lenders	Digital-first, alternative data, instant decisions	Rapid portfolio growth outpacing risk model maturity
Small Finance Banks	Financial inclusion, rural/semi-urban penetration	First-time borrowers with no credit history

4.2 Target Deployment

This tool is designed for deployment as an internal system within mid-to-large banks and NBFCs, covering retail loan underwriting (personal loans, credit cards), portfolio monitoring and early warning systems, and regulatory risk reporting under RBI and Basel III guidelines.

5. Competitor Analysis

The Indian credit information ecosystem is dominated by four licensed credit bureaus. We compare the three most relevant competitors and identify gaps addressed by this project.

5.1 Competitor Overview

Parameter	TransUnion CIBIL	CRIF High Mark	Experian India
Market Share	~55–60%	~20–25%	~10–15%
Score Range	300–900	300–900	300–900
Model Type	Proprietary logistic/scorecard	Logistic + rule engines	Ensemble (proprietary)
Explainability	Top 4 reason codes (generic)	Limited reason codes	Basic reason codes
Customisation	Limited	Some segment-specific	Industry variants

5.2 Our Differentiators

- Deep explainability: Full 13-feature SHAP waterfall with quantified per-feature impact and specific improvement recommendations, vs. 4–5 generic reason codes from bureaus.
- Calibrated P(default): Direct probability output enabling precise expected-loss calculations, rather than an opaque score requiring internal PD mapping.
- Flexible thresholds: Dual-threshold framework (Max-F1 and Precision-targeted) allowing calibration to specific risk appetites.
- Rapid retraining: Optuna-based pipeline supports automated hyperparameter search across multiple architectures (CatBoost, XGBoost, MoE).
- Cost advantage: Eliminates per-query bureau fees (INR 25–50 per query) for high-volume lenders after initial development.

6. Dataset

6.1 Dataset

The project uses the *"Give Me Some Credit"* dataset, containing 150,000 training and 101,503 test observations with 11 financial attributes.

Dataset Link: [Bank Credit Dataset](#)

Variable	Internal Name	Description	Type
SeriousDlqin2yrs	defaulted	Target: 90+ days past due within 2 years	Binary
RevolvingUtilizationOfUnsecuredLines	unsecured_credit	Total revolving balance / credit limit	Float
age	age	Borrower age in years	Integer
NumberOfTime30-59DaysPastDueNotWorse	delinq_30_59	Count of 30–59 day delinquencies	Integer
DebtRatio	debt_ratio	Monthly debt payments / gross income	Float
MonthlyIncome	monthly_income	Monthly income (INR)	Float
NumberOfOpenCreditLinesAndLoans	open_credit	Open loans and credit lines	Integer
NumberOfTimes90DaysLate	delinq_90	Count of 90+ day delinquencies	Integer
NumberRealEstateLoansOrLines	real_estate_loans	Mortgage / real estate loans	Integer
NumberOfTime60-89DaysPastDueNotWorse	delinq_60_89	Count of 60–89 day delinquencies	Integer
NumberOfDependents	dependents	Family dependents (excl. self)	Integer

7. Project Structure

7.1 Machine Learning Code: [Github Repository Link \(ML Part\)](#)

The codebase follows a modular architecture with clear separation of concerns. Each module handles a single stage of the ML pipeline.

7.1.1 Technology Stack

Category	Library / Tool	Version	Purpose
Data Processing	pandas	2.x	Data loading, cleaning, feature engineering
Data Processing	NumPy	1.26+	Array operations, numerical computation
ML — Gradient Boosting	CatBoost	1.2+	Primary production model (ordered boosting)
ML — Gradient Boosting	XGBoost	2.0+	Benchmark gradient boosting model
ML — Baseline	scikit-learn	1.4+	Logistic regression, StandardScaler, metrics
Deep Learning	PyTorch	2.x	Mixture-of-Experts neural architecture
Hyperparameter Tuning	Optuna	3.x	Bayesian TPE search (50 trials, 7 parameters)
Explainability	SHAP	0.44+	TreeExplainer for per-feature Shapley values
Visualisation	Matplotlib / Seaborn	3.8+ / 0.13+	EDA plots, confusion matrices, ROC/PR curves
Visualisation	Plotly / Altair	5.x / 5.x	Interactive UMAP projections, dashboard charts
Dimensionality Reduction	UMAP	0.5+	2D/3D borrower manifold visualisation
Deployment	Streamlit	1.30+	Interactive credit risk scoring dashboard
Integration	gsread (Google Sheets API)	6.x	Batch prediction read/write to Google Sheets
Model Persistence	Joblib / CatBoost .cbm	—	Model serialisation and metadata storage

7.1.2 Directory Layout

```
credit-risk-modelling/
├── Data/
│   ├── cs-training.csv    # 150K training records
│   ├── cs-test.csv       # 101K test records
│   └── Data Dictionary.xls # Variable descriptions
├── models/
│   ├── catboost_tuned.cbm # Tuned CatBoost model
│   └── catboost_tuned_meta.joblib # Scaler + thresholds
├── notebooks/
│   ├── eda.ipynb          # Exploratory data analysis (48 cells)
│   ├── boosting_models.ipynb # XGBoost + CatBoost training (33 cells)
│   └── moe_model_with_log_income.ipynb # PyTorch MoE (45 cells)
```

7.1.3 Module Dependency Map

Module	Imports From	Responsibility
config.py	— (standalone)	SEED, feature lists, hyperparameters, Optuna search space, file paths
data.py	config	CSV loading, sentinel treatment, missingness indicators, imputation, stratified splitting, StandardScaler
train.py	config, data	CatBoost baseline construction with class_weights and early stopping
tune.py	config, data, train	Optuna TPE search (50 trials), best-param extraction, tuned model retraining
evaluate.py	config	Threshold sweep (499 points), classification reports, ROC/PR plots, feature importance

7.2 Website Code: [Github Repository Link \(Website\)](#)

The codebase follows a modular architecture with clear separation of concerns. Each module handles a single stage of the ML pipeline, spanning from frontend customer onboarding to backend predictive diagnostics.

7.1.1 Technology Stack

Category	Library / Tool	Version	Purpose
Frontend Framework	Next.js / React	14.x+	Application shell, App Router, and UI rendering
Language	TypeScript	5.x+	Type-safe domain logic and API interfaces
Styling / UI	Tailwind / Radix UI	3.x+	Responsive CRM styling and accessible primitives
Data Processing	pandas / NumPy	2.x / 1.26+	Pipeline data loading and numerical computation
ML — Boosting	CatBoost	1.2+	Core scoring engine for risk classification
ML — Baseline	scikit-learn	1.4+	Preprocessing, pipeline utilities, and metrics

Explainability	SHAP	0.44+	Feature attribution and risk factor visualization
Integration	Googleepis	100+	CRUD operations for Google Sheets data source
Serialization	Joblib	1.3+	Model artifact and metadata persistence

7.1.2 Directory Layout

Plaintext

Credit_Scorer/

```

├── Data/
│   ├── cs-training.csv    # 150K Training records (Give Me Some Credit)
│   ├── cs-test.csv       # 101K Test records
│   └── sampleEntry.csv   # Baseline submission format
├── app/
│   ├── api/              # Next.js API Routes (Next.js Backend)
│   │   ├── customers/    # Customer CRUD + Phase 1 trigger
│   │   └── diagnose/     # Recommendation + Phase 2 trigger
│   └── page.tsx          # Main CRM Dashboard shell
├── models/
│   └── catboost_tuned_meta.joblib # Model metadata + Scaler
├── phase1_prediction.py   # Primary scoring pipeline
└── phase2_suggestions.py  # Recommendation & diagnosis engine

```

7.1.3 Module Dependency Map

Module	Imports From	Responsibility
lib/types.ts	— (standalone)	Shared TypeScript interfaces for customers and results
app/api/add	phase1_prediction.py	Appending CRM rows and triggering primary scoring
app/api/diagnose	phase2_suggestions.py	Running SHAP-based diagnosis and priority actions
phase1_prediction.py	models/, Data/	Processing raw inputs to generate risk-level scores
phase2_suggestions.py	models/	Generating actionable remediations for high-risk users
app/api/import	Google Sheets API	Bulk data ingestion to operational data source

8. Methodology

8.1 Data Preprocessing

The preprocessing pipeline is implemented in `data_preprocessing.py` and `data_loader.py`. It addresses three primary data quality issues:

- Age filtering: Records with age = 0 removed as impossible values.
- Sentinel treatment: Delinquency codes 96/98 are data-entry artefacts. A binary flag `is_delinquent` is created via `delinquency_codes`, and all delinquency counts are capped at 10 using `clip`.
- Missing value handling: Binary indicators are created before imputation. Monthly income is imputed with median (5,400); dependents filled with 0.

After preprocessing, the feature set comprises 13 variables: 10 original + 3 engineered (`age`, `is_delinquent`, `income`).

8.2 Train-Validation-Test Split

A stratified 70/15/15 split preserves the ~6.7% default rate across all partitions. Stratification is performed via `scikit-learn`. Features are standardised using `StandardScaler` fitted on the training partition only.

8.3 Class Imbalance Handling

With a class ratio of approximately 14:1 (non-default : default), class weights are applied: CatBoost uses `class_weights`, XGBoost uses `scale_pos_weight`, and the MoE model uses weighted BCE loss with `weight_decay`.

8.4 Model Architectures

8.4.1 Logistic Regression (Baseline)

Standard logistic regression provides an interpretable baseline with linear decision boundary. SHAP computes exact Shapley values. Achieved ROC-AUC of approximately 0.82 on the test set.

8.4.2 XGBoost

XGBoost (Chen & Guestrin, 2016) trained with 2,000 estimators, `learning_rate=0.1`, and early stopping at 30 rounds. Uses second-order Taylor expansion with L1/L2 regularisation. Achieved ROC-AUC of approximately 0.85.

8.4.3 CatBoost (Primary Model)

CatBoost (Prokhorenkova et al., 2018) was selected as the primary model due to its *ordered boosting* mechanism (preventing target leakage) and symmetric tree structure. Baseline parameters: . After Optuna tuning, the model achieves ROC-AUC of approximately 0.86.

8.4.4 Mixture-of-Experts (MoE)

A soft MoE model with 3 experts implemented in PyTorch. Each expert is a 3-layer MLP () with 64 hidden units. A gating network assigns soft routing weights per sample. Trained for up to 400 epochs with early stopping (patience 100) and auxiliary load-balancing loss () to prevent expert collapse. Knowledge distillation compresses the MoE into a 2-layer student network.

8.5 Threshold Calibration

Rather than using the default 0.5 threshold, 499 thresholds are swept on the validation set (in via) to identify two operating points:

- Max-F1: Threshold that maximises F1 on the default class.
- Precision $\geq 50\%$: Threshold that maximises recall subject to default-class precision $\geq 50\%$.

8.6 CIBIL Credit Score Formula

The model output is transformed into a 300–900 CIBIL-aligned score using the composite formula (implemented in):

$$Score = 300 + (Weighted_Composite \times 6)$$

$$Weighted = Default_Probability_Score \times 0.50 + Payment_History \times 0.25 + Credit_Utilisation \times 0.15 + Credit_Mix_Duration \times 0.10$$

Component	Weight	Scoring Logic
Default Probability	50%	ML model P(default) mapped to 0–100 scale
Payment History	25%	Penalises delinquencies: 90d (–25), 60d (–15), 30d (–7), sentinel (–100)
Credit Utilisation	15%	Score = max(0, 100 – utilisation × 150)
Credit Mix & Duration	10%	Age factor + open_credit × 2 + real_estate × 10

Grade	Score Range
Excellent	750 – 900
Very Good	700 – 749
Good	650 – 699
Fair	600 – 649
Poor	300 – 599

9. Model Training Pipeline

9.1 Pipeline Architecture:

End-to-End ML Pipeline for Credit Risk Scoring with Explainability and LLM Insights



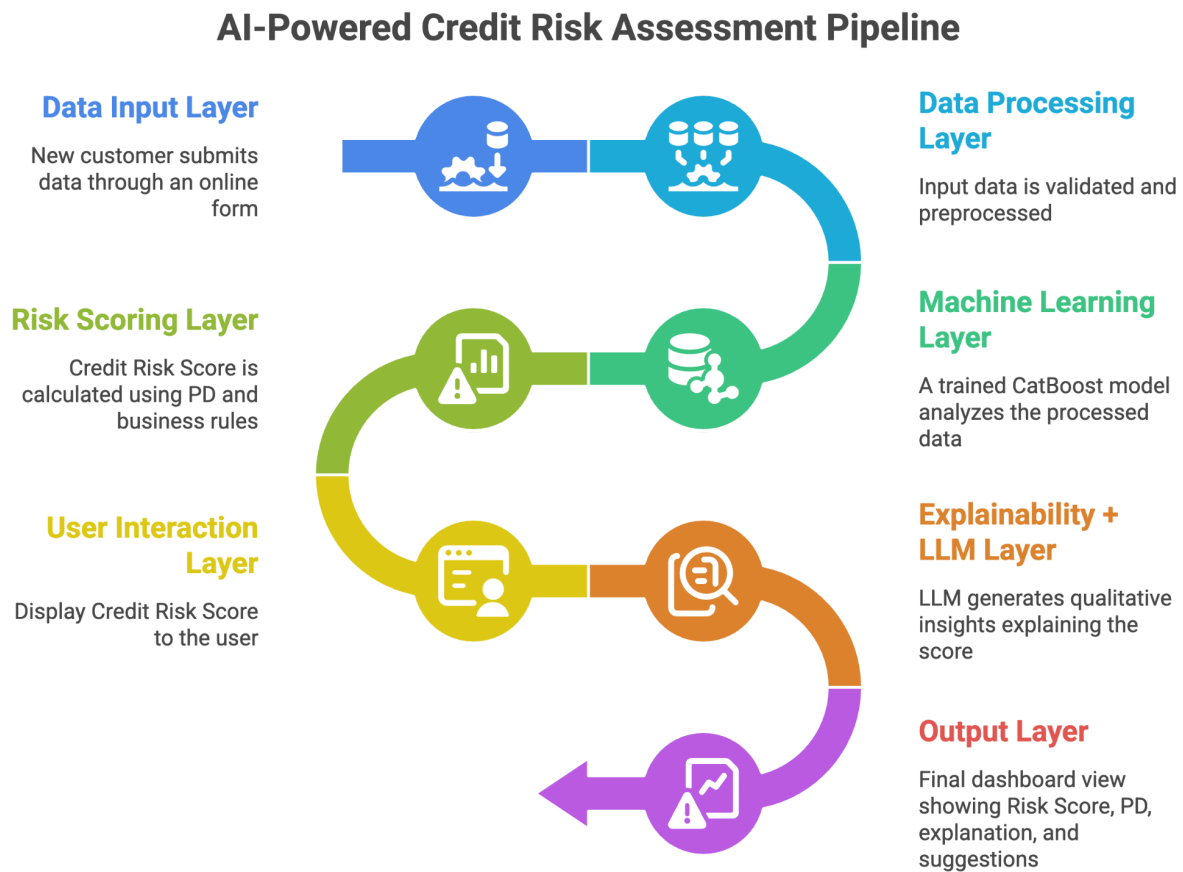
9.3 Model Comparison Summary

Model	ROC-AUC	F1-Score	Notes
Logistic Regression	~0.82		Linear baseline; exact SHAP via LinearExplainer
XGBoost	~0.85		2K estimators, depth 6; second-order gradient
CatBoost (Baseline)	~0.85		Ordered boosting; default hyperparams
CatBoost (Tuned)	~0.86		Optuna 50 trials; primary production model
MoE (3 Experts)	~0.84		Expert specialisation; knowledge distillation to student

Note: Approximate metrics are reported for illustration. Exact values depend on the random seed (SEED=42) and early stopping convergence.

10. Business Workflow and Deployment

10.1 System Architecture:



User-Journey:

1. Data Input Layer: Customer's data is submitted using this form.

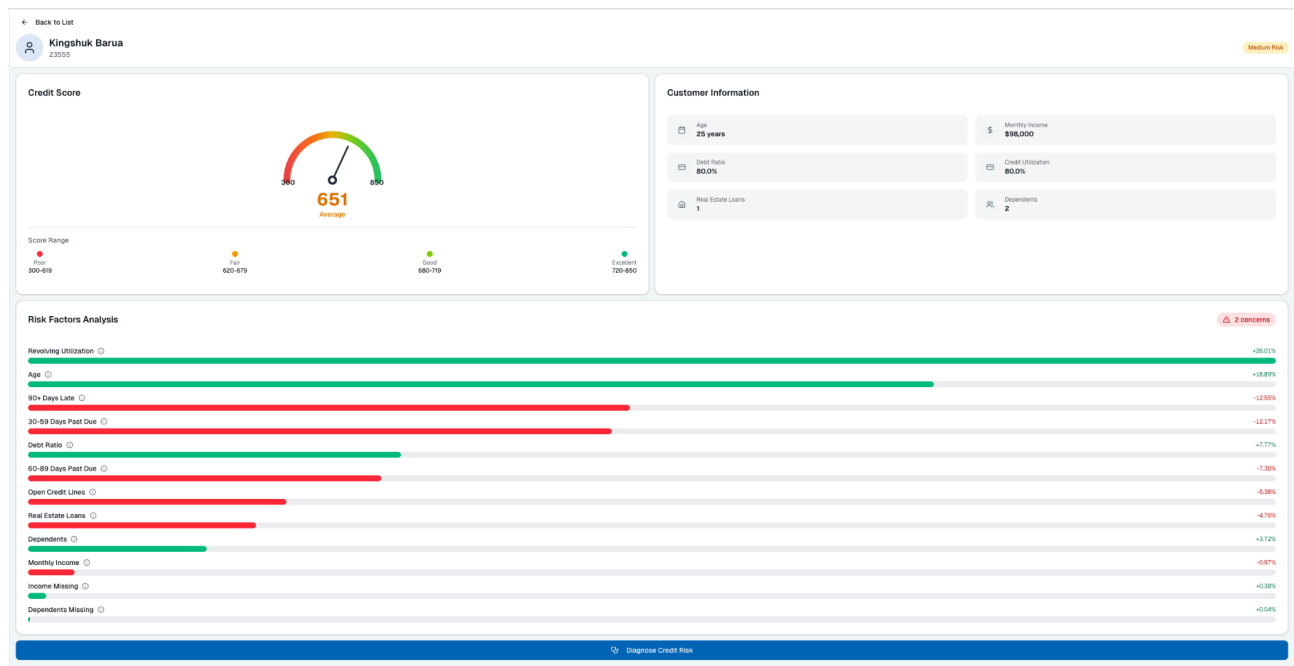
The screenshot shows the 'Credit Risk Analyzer' dashboard with a 'Customers' tab. A modal form titled 'Add New Customer' is open, allowing data entry for a new customer. The form includes fields for Monthly Income, Number of Open Credit Lines and Loans, Number of Times 90 Days Late, Number of Real Estate Loans or Lines, Number of Times 60-89 Days Past Due (Not Worse), and Number of Dependents. The background dashboard shows a list of existing customers and a detailed view for 'Kingshuk Barua' with a credit score of 651 and a medium risk level.

Customer Name	Credit Score	Risk Level	Income
Brandon Mitchell	845	Low Risk	\$50,000/mo
Charlotte Pemberton	842	Low Risk	\$4,280/mo
Divya Hegde	681	Medium Risk	\$4,000/mo
Eleanor Hartley	836	Low Risk	\$3,833/mo

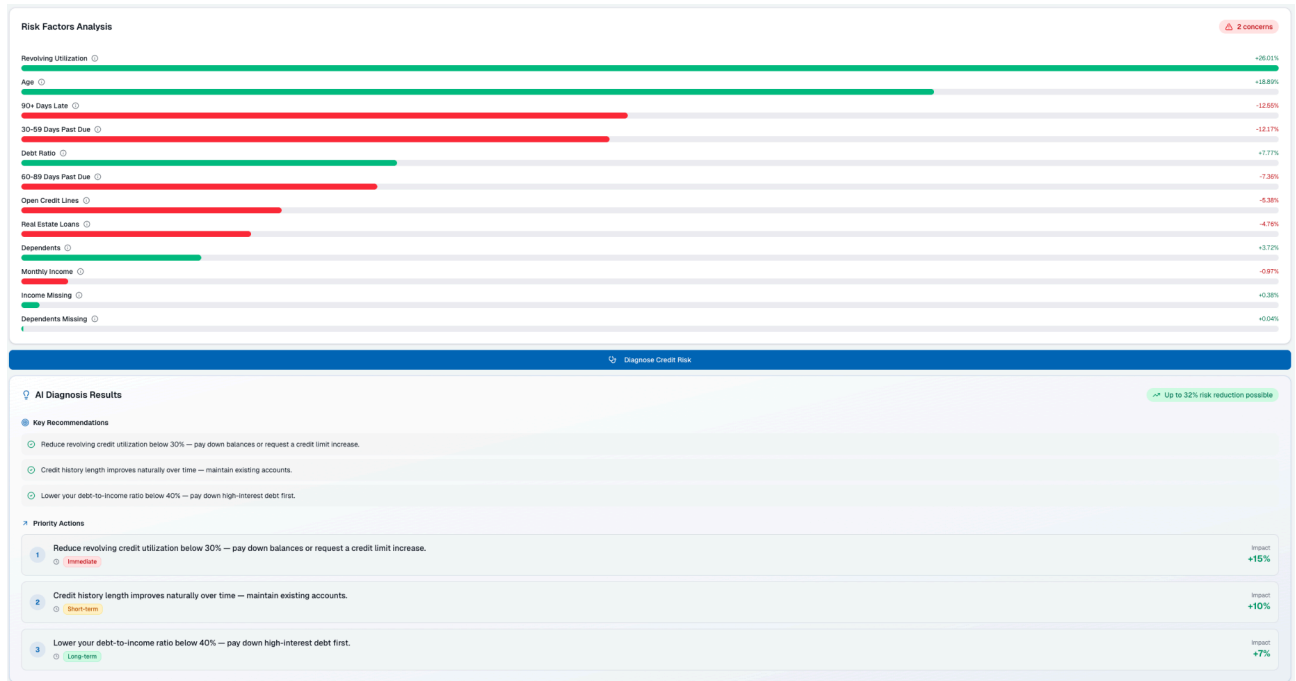
Add New Customer Form Fields:

- Monthly Income: 5400
- Number of Open Credit Lines and Loans: 1
- Number of Times 90 Days Late: 0
- Number of Real Estate Loans or Lines: 1
- Number of Times 60-89 Days Past Due (Not Worse): 0
- Number of Dependents: 0

2. User Interaction Layer: Credit profiling details of the newly added user is displayed.



3. Diagnose Credit Risk:



12. Key Project Insights

12.1 Model Performance

- CatBoost outperforms XGBoost marginally: Ordered boosting provides a slight edge in ROC-AUC (~0.86 vs. ~0.85), likely due to superior handling of gradient estimation leakage.
- Optuna tuning yields diminishing returns: Tuned model improves ~0.5–1% in AUC over baseline. fANOVA analysis reveals and are the most impactful hyperparameters.
- MoE expert specialisation is meaningful: Three experts specialise on distinct borrower segments with different default rates, confirmed by UMAP projections showing clear expert territories.

12.2 Feature Importance

- Delinquency features dominate: Across all models, , , and are consistently the top predictors.
- Revolving utilisation is the top non-delinquency feature: High (balance / credit limit) is strongly associated with default.

- Missingness is informative: carries significant predictive power. Borrowers with missing income have ~9.7% default rate vs. ~6.0% for those with income reported.

12.3 Business Insights

- Threshold choice matters more than model choice: The difference between Max-F1 and Precision-targeted thresholds has a far larger impact on business outcomes than model architecture differences.
- SHAP enables targeted interventions: For borderline cases, SHAP identifies the one or two features that, if improved, would flip the prediction — enabling conditional approvals.
- Score compositing adds value: Transforming raw probability into a 300–900 CIBIL-aligned score makes the output immediately interpretable by loan officers trained on bureau scores.

13. Limitations and Future Work

13.1 Current Limitations

Limitation	Impact	Mitigation Strategy
Static dataset ()	Does not reflect real-time Indian credit dynamics	Retrain on live institutional data when available
No temporal modelling	Payment behaviour trajectories not captured	Add LSTM/Transformer sequence models over time windows
No alternative data	UPI transactions, GST filings, social signals not used	Integrate alternative data APIs for thin-file borrowers
Single dataset validation	Generalisability to Indian borrower profiles unverified	Cross-validate on Indian bureau data samples
Approximate CIBIL mapping	Score formula is illustrative, not calibrated to CIBIL proprietary methodology	Calibrate with actual CIBIL score distributions from partner banks
No fairness auditing	Potential demographic bias in predictions unchecked	Add Aequitas / AI Fairness 360 bias auditing pipeline
No model monitoring	No drift detection or performance decay alerting	Integrate Evidently AI or custom PSI/CSI monitoring

13.2 Future Work

- Temporal modelling: Sequence-based architectures (LSTM, Transformer) to capture payment behaviour trajectories and early warning signals over time.
- Alternative data integration: UPI transaction patterns, GST filings for MSME borrowers, telecom/utility payment history for thin-file populations.
- Federated learning: Multi-institution model training without sharing raw borrower data, enabling collaborative risk modelling across banking networks.
- Real-time API deployment: FastAPI/Flask wrapper with Redis caching for sub-100ms inference latency in production lending workflows.
- Regulatory compliance: Full model risk management (MRM) documentation per RBI guidelines, including model validation report, stress testing, and backtesting framework.

13.3 Known Technical Debt

- Hardcoded median: `in` should be loaded from model metadata for consistency across retraining cycles.
- Sequential Sheet writes: writes one row at a time; `batch` calls would reduce API round-trips.
- No unit tests: The codebase lacks a formal test suite. Adding `pytest` coverage for preprocessing, prediction, and CIBIL scoring functions is recommended.

14. Conclusion

This project demonstrates a complete, production-ready credit risk modelling framework that bridges the gap between advanced machine learning and practical credit decisioning. By combining CatBoost gradient boosting with Optuna hyperparameter optimisation, a neural Mixture-of-Experts architecture, and SHAP-based explainability, the system achieves both predictive accuracy (ROC-AUC ~0.86) and interpretability.

The key contributions are: (a) a dual-threshold decision framework enabling precision-recall calibration to institutional risk appetite; (b) a CIBIL-aligned credit scoring formula weighting ML-derived default probability at 50%, integrating predictions into the familiar 300–900 range; (c) per-prediction SHAP explanations with actionable improvement recommendations supporting fair lending and customer engagement; and (d) a deployment-ready system with a Streamlit dashboard and Google Sheets batch integration.

The economic analysis suggests a mid-size bank deploying this tool internally could realise annual savings exceeding INR 24 crore through reduced credit losses, bureau fee substitution, and operational efficiency, with a payback period under 2 months.

The framework is designed for extensibility. Future iterations incorporating temporal modelling, alternative data sources, and federated learning would further enhance its applicability across the Indian banking ecosystem.

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